Solving Radical Equations and Checking Extraneous Roots Answer Key

Algebra 2

Quick Answers

1. 28	4. 5, 0, 5, 2	7. , 6	10. 4, -1, 4, 4
2. 8	5. 100	8. 4	
3. -2, 3, 0, 4	6. 150	9. 9	

1. Solve $\sqrt{x-3}=5$.
Solution: the radical is already isolated, so square both sides: $x-3=25$, hence $x=28$. Check: $\sqrt{28-3}=\sqrt{25}=5$ ✓. Both sides were non-negative before squaring, so no extraneous root could appear. $\boxed{x=28}$

2. Rewrite and evaluate $16^{3/4}$.
Solution: a rational exponent is root-then-power: $16^{3/4}=\left(\sqrt[4]{16}\right)^3=2^3=8$. Taking the fourth root first keeps the arithmetic small; the other order needs $16^3=4096$ before rooting. $\boxed{8}$

3. $\sqrt{x+6}=x$, squared equation $x^2-x-6=0$.
(a) $x^2-x-6=(x-3)(x+2)=0$ gives the candidates $\boxed{x=-2}$ and $\boxed{x=3}$.
(b) Substituting into $\sqrt{x+6}-x$: at $x=3$: $\sqrt{9}-3=0$ ✓; at $x=-2$: $\sqrt{4}-(-2)=2+2=\boxed{4}\neq 0$ ×.
(c) $x=-2$ is **extraneous** — it solves $x+6=x^2$ but makes the original read $2=-2$, and the principal square root is never negative. Solution: $\boxed{x=3}$

4. Solve $\sqrt{3x+1}=x-1$.
(a) Squaring: $3x+1=x^2-2x+1$, so $x^2-5x=0$ and $x(x-5)=0$, giving candidates $\boxed{x=0}$ and $x=5$.
(b) At $x=0$: $\sqrt{3(0)+1}-(0-1)=1+1=\boxed{2}\neq 0$ × — the left side is 1 while the right side is -1, so this candidate is extraneous. At $x=5$: $\sqrt{16}=4$ and $5-1=4$ ✓.
(c) Solution set: $\boxed{x=5}$
Why it happened: the right side $x-1$ is negative at $x=0$, and a square root can never equal a negative number.

5. Free fall: $t=\sqrt{h/16}$ with $t=2.5$ seconds; find h in feet.
Solution: substitute and solve the radical equation:

$$2.5=\sqrt{\frac{h}{16}} \implies 6.25=\frac{h}{16} \implies h=100$$

Check in the model: $\sqrt{100/16}=\sqrt{6.25}=2.5$ ✓. Both sides are non-negative here (a time and a principal root), so the candidate cannot be extraneous. $\boxed{h=100\text{ ft}}$

6. Skid marks: $s = \sqrt{24x}$ with $s = 60$ miles per hour; find x in feet.
Solution:

$$60 = \sqrt{24x} \implies 3600 = 24x \implies x = 150$$

Check: $\sqrt{24 \cdot 150} = \sqrt{3600} = 60$ ✓. $\boxed{x = 150 \text{ ft}}$
Domain note: only $x \geq 0$ makes sense as a skid length, and the model's own square root already refuses negative inputs.

7. Solve $\sqrt{x-2} = -3$, or explain why there is no real solution.
Solution: squaring gives $x - 2 = 9$, so $x = 11$ is the only candidate. Substituting into $\sqrt{x-2} + 3$ at $x = 11$: $\sqrt{9} + 3 = \boxed{6} \neq 0$, i.e. $\sqrt{9} = 3 \neq -3$ ×.
The principal square root outputs only values ≥ 0, so $\sqrt{x-2} = -3$ is impossible for every real x; the candidate is extraneous and the equation has $\boxed{\text{no real solution}}$
This is the cleanest illustration of why the check is mandatory: squaring manufactured a root out of an equation with none.

8. Solve $x^{3/2} = 8$.
Solution: in radical form the equation is $(\sqrt{x})^3 = 8$. Take the cube root of both sides: $\sqrt{x} = 2$, then square: $x = 4$. Check: $4^{3/2} = \left(\sqrt{4}\right)^3 = 2^3 = 8$ ✓. $\boxed{x = 4}$
No negative solution: $x^{3/2}$ contains $\sqrt{x}$, which is undefined for $x < 0$ in the real numbers — the domain rules negatives out before any solving happens.

9. Solve $\sqrt{x+7} - \sqrt{x} = 1$.
Solution: isolate one radical first, or squaring leaves two of them:

$$\begin{aligned}
\sqrt{x+7} &= 1 + \sqrt{x} && \\
x + 7 &= 1 + 2\sqrt{x} + x && \text{(square both sides)} \\
6 &= 2\sqrt{x} \implies \sqrt{x} = 3 && \text{(isolate the radical)} \\
x &= 9 && \text{(square again)}
\end{aligned}$$

Check: $\sqrt{9+7} - \sqrt{9} = 4 - 3 = 1$ ✓. $\boxed{x = 9}$

10. $\sqrt{x+5} = x - 1$ with the table of both sides.
(a) At $x = -1$: $\sqrt{4} = 2$ but $x - 1 = -2$ — *not equal.* At $x = 4$: $\sqrt{9} = 3$ and $x - 1 = 3$ — *equal.* (Squaring gives $x + 5 = x^2 - 2x + 1$, i.e. $x^2 - 3x - 4 = 0$, whose roots are the two candidates $\boxed{x = -1}$ and $x = 4$.)
(b) Solution set: $\boxed{x = 4}$
(c) The curve $y = \sqrt{x+5}$ lies on or above the x-axis everywhere, so it can only meet $y = x - 1$ where that line is non-negative, i.e. for $x \geq 1$ — one intersection. Squaring replaces the line by the parabola $y = (x-1)^2$, which is non-negative on both sides of $x = 1$, so it also meets the graph of $y = x + 5$ at $x = -1$. The table shows the cost: at $x = -1$ the two original sides differ by $\boxed{4}$, so that intersection belongs to the squared equation only.